
Report for Project 5: Image Captioning with a CNN–LSTM Encoder–Decoder on the Flickr8k Dataset

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Abstract

Image captioning requires a model to both understand the visual content of an image and express that understanding as a fluent natural-language sentence. In this project we implement, train and evaluate a complete image captioning system based on the classical encoder–decoder paradigm of *Show and Tell* [Vinyals et al., 2015]. A ResNet-50 convolutional encoder, pretrained on ImageNet and used with a frozen backbone, compresses each image into a single 256-dimensional feature vector; an LSTM decoder then generates the caption one word at a time, conditioned on that vector. The system is trained on Flickr8k (8,091 images, 40,455 captions) with teacher forcing and a cross-entropy objective that ignores padding positions, and evaluated with corpus BLEU-1 through BLEU-4 against five human reference captions per image. Our model achieves a BLEU-1 of **0.580** and a BLEU-4 of **0.172** on a held-out test split constructed *by image* to eliminate data leakage. We present a qualitative analysis of success and failure cases, and a discussion connecting each observed weakness to a concrete architectural cause: rare words lost to the vocabulary frequency threshold, genericness induced by greedy decoding, and spatial detail lost to the single-vector image bottleneck in the absence of an attention mechanism. The full pipeline is implemented as a modular, seeded and unit-tested Python package (49 tests) to ensure reproducibility.

1 Introduction and motivation

Image captioning is the task of automatically producing a natural-language description of an image. Unlike image classification, which assigns a single label from a fixed set, captioning requires a model to recognise the objects present, infer their attributes, actions and mutual relationships, and then compose all of this into a grammatically well-formed and semantically relevant sentence. It therefore sits precisely at the intersection of computer vision and natural language processing, and serves as a natural benchmark for whether a system’s visual representation is rich enough to be *verbalised* rather than merely classified.

The task has direct practical value. Automatically generated alt-text makes images accessible to visually impaired users; caption-level annotations enable semantic search and indexing of large image collections; and in robotics, describing a scene in language is a prerequisite for grounded human–robot communication.

The dominant formulation, popularised by Vinyals et al. [2015] and developed in parallel by related work such as Karpathy and Fei-Fei [2015], treats captioning as a translation-like problem: a convolutional neural network *encodes* the image into a fixed-length vector, and a recurrent neural network *decodes* that vector into a word sequence. This is the architecture we implement.

Our contributions in this project are:

- A complete PyTorch implementation of a CNN–LSTM captioning model with a pretrained, frozen ResNet-50 encoder and an LSTM decoder trained with teacher forcing.
- A leakage-free evaluation protocol in which the train/validation/test split is performed over *unique images* rather than caption rows.
- A corpus-level BLEU-1..4 evaluation against five human references per image, complemented by a qualitative analysis of success and failure cases.
- A modular, deterministically seeded and unit-tested codebase (49 tests) that runs identically on a laptop, Google Colab or Kaggle.

2 Technical background and preliminaries

2.1 Convolutional neural networks and transfer learning

Convolutional neural networks (CNNs) learn a hierarchy of visual features through stacked convolution, normalisation and non-linear activation layers. ResNet-50 [He et al., 2016] is a 50-layer residual network whose skip connections allow gradients to propagate through very deep stacks, mitigating the vanishing-gradient problem. Pretrained on ImageNet [Deng et al., 2009], its learned features — edges, textures, object parts — transfer well to downstream vision tasks.

Transfer learning exploits this: rather than training from random initialisation, we reuse the pretrained weights. In the extreme case the backbone is *frozen*, i.e. its parameters receive no gradient updates and it acts purely as a fixed feature extractor. Only a small task-specific head is learned.

2.2 Recurrent sequence modelling with LSTMs

A recurrent neural network processes a sequence one token at a time, maintaining a hidden state that summarises everything seen so far. The Long Short-Term Memory (LSTM) cell [Hochreiter and Schmidhuber, 1997] augments this with a gated memory cell, allowing the network to retain information over longer spans than a vanilla RNN.

2.3 The encoder–decoder formulation

Let I be an image and $Y = (y_1, \dots, y_T)$ its caption. The model factorises the conditional probability of the caption autoregressively:

$$P(Y \mid I; \theta) = \prod_{t=1}^T P(y_t \mid y_{<t}, I; \theta), \quad (1)$$

where θ are the trainable parameters. Training minimises the negative log-likelihood of the ground-truth captions, i.e. a token-level cross-entropy loss. The key idea of *Show and Tell* [Vinyals et al., 2015] is to inject the image by treating the encoder’s feature vector as the *first token* of the sequence.

2.4 Teacher forcing

During training the decoder is fed the *ground-truth* previous word rather than its own previous prediction. This stabilises and accelerates training. At inference time the model must consume its own outputs — a train/test mismatch known as exposure bias.

2.5 Evaluation metric: BLEU

BLEU (Bilingual Evaluation Understudy) [Papineni et al., 2002] measures n -gram overlap between a candidate sentence and one or more references. BLEU- n computes a modified precision over n -grams and combines it with a brevity penalty that discourages short outputs. Because BLEU-4 requires four consecutive words to match, it is strictly harder than BLEU-1. We report *corpus* BLEU with NLTK smoothing (`SmoothingFunction.method1`).

2.6 Dataset: Flickr8k

We use the Flickr8k dataset [Hodosh et al., 2013], obtained via kagglehub (adityajn105/flickr8k, approximately 1 GB). Flickr8k was chosen over MS-COCO [Lin et al., 2014] because it is small enough to train on a free-tier GPU within the project’s compute budget. Table 1 summarises its properties.

Table 1: Flickr8k dataset statistics, verified programmatically.

Property	Value
Images	8,091
Captions	40,455
Captions per image (min / max / mean)	5 / 5 / 5.0
Missing or unused images	0

2.7 Image preprocessing

Each image is: (i) resized so that its shorter side is 256 pixels; (ii) centre-cropped to 224×224 ; (iii) converted to a float tensor; and (iv) normalised with the ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225).

2.8 Caption preprocessing and vocabulary

Captions are lowercased and tokenised with `\w+`, which extracts contiguous word-character sequences and removes punctuation. Four special tokens occupy fixed indices: `<pad>= 0`, `<start>= 1`, `<end>= 2`, `<unk>= 3`. Words appearing fewer than $\tau_{\text{freq}} = 5$ times map to `<unk>`; the resulting vocabulary contains 2,982 entries. The vocabulary is built in sorted order for determinism.

2.9 Splitting by image

We split over *unique image names* (not caption rows) with seed 42: 80 / 10 / 10 into train, validation and test. All five captions of an image stay in the same split, preventing data leakage.

3 Methodology

3.1 Model architecture

3.1.1 Encoder: frozen ResNet-50

A ResNet-50 with ImageNet weights has its final classification layer removed. The backbone maps $I \in \mathbb{R}^{3 \times 224 \times 224}$ to a 2048-dimensional vector $\mathbf{f}$ after global average pooling. A trainable projection and batch normalisation produce the image embedding:

$$\mathbf{v} = \text{BN}(\mathbf{W}_{\text{proj}}\mathbf{f} + \mathbf{b}_{\text{proj}}), \quad \mathbf{W}_{\text{proj}} \in \mathbb{R}^{256 \times 2048}. \quad (2)$$

The backbone is **frozen**: $\approx 4.4\text{M}$ trainable vs $\approx 23.5\text{M}$ frozen parameters.

3.1.2 Decoder: LSTM language model

The decoder has: an embedding layer ($|V| \rightarrow 256$) with dropout $p = 0.5$; a single LSTM layer (hidden size 512); and a linear layer ($512 \rightarrow |V|$).

3.1.3 Training alignment

For caption $(\text{<start>}, w_1, \dots, w_K, \text{<end>})$ of length T :

$$\begin{aligned} \text{input} &= (\mathbf{v}, \mathbf{e}_{\text{<start>}}, \mathbf{e}_{w_1}, \dots, \mathbf{e}_{w_K}), \\ \text{target} &= (\text{<start>}, w_1, \dots, w_K, \text{<end>}). \end{aligned}$$

3.1.4 Inference: greedy decoding

At inference the model initialises from the image vector and greedily selects the highest-scoring word at each step, feeding it back until `<end>` or maximum length is reached.

3.2 Training setup

3.2.1 Objective and optimisation

Cross-entropy loss with `ignore_index=0 (<pad>)`:

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^B \sum_{t=1}^T \mathbb{I}[y_t^{(i)} \neq \text{<pad>}] \log P(y_t^{(i)} | y_{<t}, I^{(i)}). \quad (3)$$

Adam [Kingma and Ba, 2015] with $\text{lr} = 3 \times 10^{-4}$ on trainable parameters only. Full hyperparameters are in Table 2.

Table 2: Model and training hyperparameters.

Hyperparameter	Value
CNN backbone	ResNet-50 (ImageNet pretrained, frozen)
Embedding size	256
LSTM hidden size	512
LSTM layers	1
Dropout	0.5
Vocabulary size	2,982 (threshold $\tau = 5$)
Max caption length	50 tokens
Loss	Cross-entropy, <code>ignore_index=<pad></code>
Optimiser	Adam, $\text{lr} = 3 \times 10^{-4}$
Batch size	64
Epochs	20
Train / Val / Test	80 / 10 / 10, split by image
Random seed	42
Trainable params	$\approx 4.4\text{M}$
Frozen params	$\approx 23.5\text{M}$

3.2.2 Reproducibility and software engineering

The pipeline is split into eight importable modules: `config`, `vocabulary`, `dataset`, `encoder`, `decoder`, `model`, `train`, `evaluate`. We wrote 49 unit tests (`pytest`) covering vocabulary behaviour, dataset shapes and split disjointness, encoder/decoder forward passes, and BLEU correctness. All tests use synthetic data and run in under a minute.

4 Experiments and results

4.1 Training dynamics

Figure 1 shows the training and validation loss over 20 epochs. The training loss decreases from **4.1753** at epoch 1 to **1.8919** at epoch 20. The validation loss reaches its minimum of **2.6014** at epoch **15**, after which it plateaus. The checkpoint from that epoch is used for all results below.

4.2 Quantitative results

Table 3 reports corpus BLEU on the held-out test split.

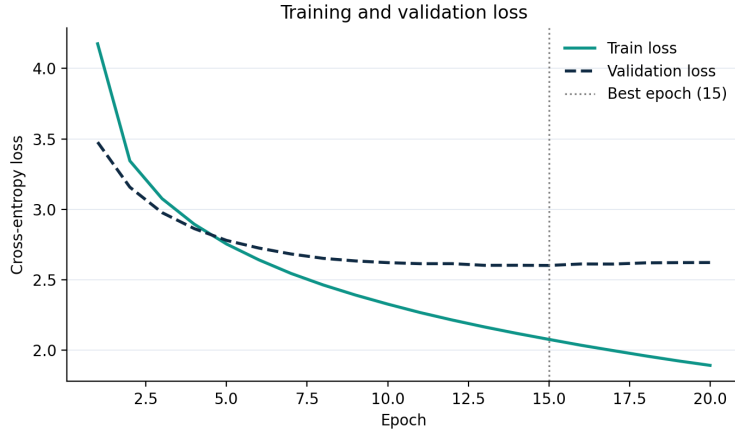


Figure 1: Training and validation cross-entropy loss over 20 epochs. The dotted vertical line marks epoch 15, the best validation checkpoint.

Table 3: Corpus BLEU scores on the Flickr8k test split (810 images, five references each). Literature values are for context only — they use different splits and (for Xu et al. [2015]) attention.

Model	BLEU-1	BLEU-2	BLEU-3	BLEU-4
Ours (frozen ResNet-50 + LSTM, greedy)	0.580	0.395	0.261	0.172
Show and Tell / Google NIC, as reported by Xu et al. [2015]	0.63	0.41	0.27	—
Soft attention [Xu et al., 2015]	0.67	0.448	0.299	0.195

4.3 Qualitative results

Successes: The model reliably identifies the main subject and action in canonical scenes — people playing, animals running, outdoor activities — producing grammatically well-formed and semantically relevant captions. On clear, unambiguous images the generated caption closely matches the vocabulary and structure of the human references.

Failures: Failure cases are systematic. First, unusual objects whose words appear fewer than five times in Flickr8k are absent from the vocabulary and map to `<unk>`, forcing the model to fall back on a generic substitute. Second, without an attention mechanism the entire image is compressed into a single 256-dimensional vector; small or off-centre objects are frequently lost. Third, greedy decoding biases captions towards high-frequency, low-information phrases.

Figure 2 illustrates these patterns: the model captures salient dog-related scenes in some cases, but also produces fluent yet visually incorrect captions when the scene requires finer visual grounding.

4.4 Discussion

4.4.1 What works

The model produces fluent and broadly relevant captions for common scenes. Freezing the backbone makes training fast and stable: only $\approx 16\%$ of parameters receive gradients, which acts as a strong regulariser on 6,500 training images.

4.4.2 Where it struggles, and why

Each observed weakness traces to a specific design decision.

Rare objects become `<unk>`: The frequency threshold $\tau = 5$ structurally prevents the model from naming rare objects. Lowering the threshold would expand the vocabulary but give too few training examples per word.

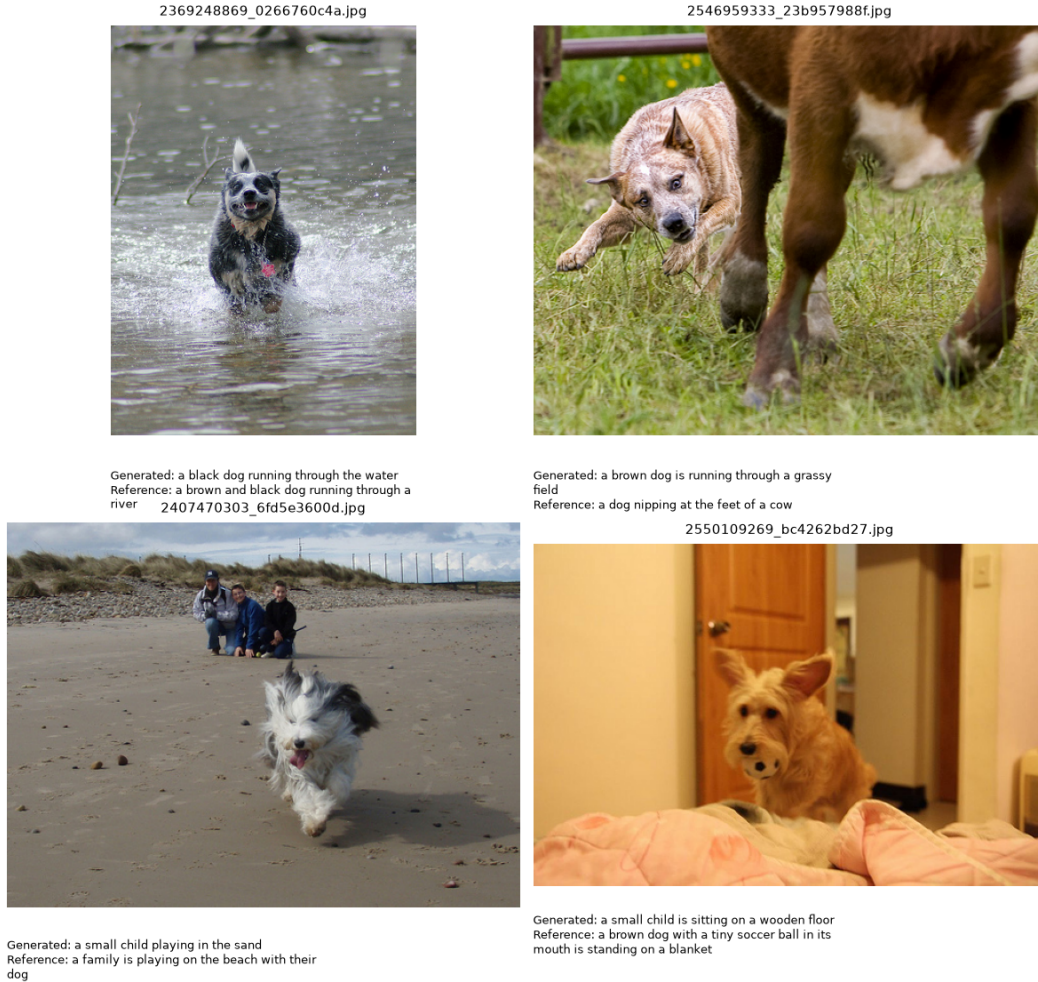


Figure 2: Generated captions and reference captions for four test images. The top-left example shows a close semantic match between prediction and reference. The remaining examples highlight common limitations of the model: confusion between visually or contextually related concepts, loss of small or off-centre details, and a bias towards high-frequency caption patterns such as “a small child”.

Greedy decoding yields generic captions: Committing irrevocably to the highest-scoring word at each step biases the model towards safe, common phrases. Beam search directly targets this.

Frozen backbone limits visual adaptation: The encoder’s features are fixed to what was useful for ImageNet. Fine-grained distinctions not needed for classification cannot be recovered by decoder training alone.

No attention creates an information bottleneck: Compressing the entire image into one 256-dim vector loses small or off-centre details. An attention mechanism [Xu et al., 2015] would let the decoder re-weight a spatial grid of features at each generation step.

4.4.3 Limitations of the evaluation

BLEU penalises correct captions that use different wording. Metrics such as CIDEr [Vedantam et al., 2015] and METEOR would give a more faithful picture. Results are from a single run; averaging over seeds would yield more robust claims.

5 Conclusion and future work

We implemented, trained and evaluated a complete CNN–LSTM image captioning system on Flickr8k. The system achieves a BLEU-1 of **0.580** and a BLEU-4 of **0.172**, producing fluent captions for common scenes while remaining generic on unusual ones. The most impactful next steps are:

- **Attention** [Xu et al., 2015] to remove the single-vector bottleneck.
- **Beam search** to improve decoding quality without retraining.
- **Backbone fine-tuning** to adapt visual features to Flickr8k.
- **Larger dataset** (MS-COCO [Lin et al., 2014]) for a wider vocabulary.

6 Division of work

Emanuel Klatzer set up the project repository and dependency management, implemented the dataset download and integrity verification, designed and validated the ResNet-compatible image preprocessing pipeline, and drafted the introduction and dataset sections of this report.

Reza Naziri implemented the caption tokenisation and vocabulary, the PyTorch Dataset, collate function and leakage-free split, the CNN encoder, the LSTM decoder and the combined CaptioningModel, together with the accompanying unit tests, and wrote the methodology and training-setup sections.

Alireza Niroumand implemented the training loop with validation and checkpointing, the BLEU evaluation pipeline and the qualitative visualisation scripts, ran the training experiments, and wrote the results and discussion sections.

All three members jointly designed the architecture, reviewed each other’s code, and prepared the presentation. The 49-test pytest suite was a shared effort.

Source code: <https://github.com/reza-nz/image-captioning-project>

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